SIX SIGMA DMAIC REPORT

CMO Response Time Reduction & Incident Management Automation

Sanitized portfolio case study | Define - Measure - Analyze - Improve - Control

Project	Business Objective	Methodology	Scope
AI-Powered CMO Response Time Reduction & Incident Management Automation System	Improve incident response, operational visibility and CMO-related performance	Six Sigma DMAIC + Agile Delivery	Reactive incident automation and proactive critical incident monitoring

1. Executive Summary

This report documents a Six Sigma DMAIC initiative focused on improving telecom incident response and operational visibility. The baseline process depended on manual monitoring of incident updates, manual stakeholder routing, manual follow-up and fragmented reporting. The improvement combined two integrated automation modules: Reactive Incident Automation and Proactive Critical Incident Monitoring with dashboards.

Measure	Before	After	Verified Outcome
Total CMO-related incident dockets	607	494	19% reduction
30-60 minute dockets	71	58	18% reduction
1-2 hour dockets	161	104	35% reduction
Dockets open for more than 3 hours	248	202	19% reduction

Evidence note: Outcome values are based on the operational comparison data provided for the project. The report attributes the improvement to better response, tracking, escalation and visibility; it does not claim that automation directly prevented underlying network faults.

2. DEFINE

2.1 Problem Statement

Network incident updates were monitored and acted on through a manual process. Raw and unstructured alerts required individual interpretation before reaching the relevant team. This created delayed ownership, inconsistent follow-up, late escalation, high manual effort and limited leadership visibility, increasing exposure to CMO and SLA risk.

2.2 Project Goal

Reduce CMO-related incident dockets and improve response effectiveness by implementing standardized, AI-powered alert processing, ownership routing, follow-up, escalation and operational KPI visibility.

2.3 Voice of Customer (VOC) and Critical to Quality (CTQ)

Customer / Stakeholder	Voice of Customer	CTQ Requirement
Fiber Operations Team	"Critical incidents should reach the right owner quickly with clear details."	Timely structured notification; correct ownership mapping.
Area Managers	"Open incidents need consistent follow-up and clear escalation."	Reminder cadence, visible duration and threshold-based escalation.
Operations Leadership	"We need a current view of open risks and performance."	Near real-time dashboard, KPI trend and refresh visibility.
Automation Support	"Exceptions must be traceable and safe to investigate."	Audit logs, exception handling and protected configuration.

2.4 SIPOC

Suppliers	Inputs	Process	Outputs	Customers
NOC and approved operational sources	Incident updates, status updates, restoration updates	Capture -> Validate -> Classify -> Notify -> Follow Up -> Escalate -> Close -> Report	Structured alerts, reminders, escalations, dashboard KPIs, audit logs	Fiber Operations, Area Managers, Operations Leadership, Support Users

2.5 Project Charter Summary

- In scope: alert processing, ownership mapping, reminders, escalation, restoration lifecycle, critical-incident monitoring and dashboard reporting.
- Out of scope: network fault resolution, field maintenance, core network configuration and infrastructure upgrades.
- Primary stakeholders: Operations Leadership, Fiber Operations, NOC Team, Area Managers, Circle Operations Head and Project Lead.

3. MEASURE

3.1 Baseline Process Measures

Metric	Baseline Condition	Measurement Method
CMO-related incident dockets	607 total dockets in the baseline comparison period	Operational docket count comparison
30-60 minute duration band	71 dockets	Duration-band comparison

Metric	Baseline Condition	Measurement Method
1-2 hour duration band	161 dockets	Duration-band comparison
More than 3 hour duration band	248 dockets	Duration-band comparison
Incident visibility	Manual / delayed	Process assessment
Follow-up and escalation	Manual / inconsistent	Process assessment
Leadership reporting	Manual compilation	Process assessment

3.2 Data Collection Plan

Data Item	Source	Frequency	Owner / Validation
Incident reference, status and timestamp	Approved operational notification source	Event-driven	Operations user validates exceptions
Open incident duration and escalation status	Automation lifecycle log	Near real-time / scheduled	Project Lead and operations users
CMO-related docket counts	Operational performance data	Comparison period	Operations Leadership review
Dashboard refresh status	Reporting process	Each refresh	Automation support owner

3.3 Measurement Interpretation

The primary outcome measure is overall CMO-related incident dockets. Duration-band figures provide supporting evidence of response and tracking improvement. One duration band may vary period-to-period; therefore, the report focuses on the total evidenced reduction and clearly reports the individual bands rather than claiming every category improved uniformly.

4. ANALYZE

4.1 Root Cause Analysis (Fishbone Summary)

Cause Category	Observed / Potential Causes	Effect on CMO Response
People	Manual interpretation, individual availability, inconsistent handoffs and follow-up ownership.	Delay before action; inconsistent status updates.
Process	No standardized alert lifecycle, reminder schedule or escalation trigger.	Long-running incidents may not receive timely attention.
Technology	Unstructured messages, limited automation, periodic extraction and fragmented reporting.	Slow visibility; higher manual workload.
Data	Incomplete notifications, varying formats, missing identifiers or owner mappings.	Routing errors or manual-review dependency.
Measurement	No single near real-time view of pending incidents, duration and performance.	Leadership sees risk late; trends are harder to act on.

4.2 Five Whys: Delayed Response to Critical Incidents

Why	Finding
1. Why are critical incidents acted on late?	Relevant teams may not receive a structured alert promptly.
2. Why is routing not prompt?	A user must manually read, interpret and forward raw notifications.
3. Why is manual interpretation required?	Incoming alerts vary in structure and are not consistently classified or mapped.
4. Why are reminders and escalation inconsistent?	Follow-up depends on individual judgement and availability.
5. Why is leadership visibility delayed?	Status and reporting are manually compiled rather than continuously aggregated.

4.3 Failure Mode and Effects Analysis (FMEA)

Failure Mode	Effect	S	O	D	RPN	Recommended Control
Critical alert not routed	Delayed ownership and response	9	4	5	180	Validation, mapping review and processing logs
Duplicate alert flood	User fatigue and reduced trust	5	5	4	100	Duplicate suppression rules
Missing owner mapping	No automated notification	8	3	4	96	Manual-review queue and mapping maintenance
Escalation not triggered	Long-running incident not highlighted	9	3	4	108	Threshold monitoring and alert logs
Dashboard data is stale	Leadership decision based on outdated view	6	3	5	90	Refresh timestamp and health monitoring

FMEA scoring convention: Severity (S), Occurrence (O) and Detection (D) are indicative 1-10 portfolio scores used to prioritize controls; they are not a claim of a formally audited production FMEA.

5. IMPROVE

5.1 Improvement Design

Improvement Component	Process Gap Addressed	Improvement Mechanism
Reactive Incident Automation	Manual interpretation and routing	Capture approved alerts, validate fields, classify criticality and send structured notifications to mapped owners.

Improvement Component	Process Gap Addressed	Improvement Mechanism
Automated Follow-up and Escalation	Inconsistent status chasing and late escalation	Apply approved intervals and duration/severity thresholds to eligible open incidents.
Restoration Lifecycle	Unclear closure communication	Process valid restoration updates, notify stakeholders and update incident status.
Proactive Critical Incident Monitoring	Periodic, manual monitoring of critical open items	Consolidate pending incidents by area and status for timely intervention.
Executive KPI Dashboard	Manual reporting and delayed leadership visibility	Publish CMO-related performance, open risk and operational KPI views with refresh visibility.

5.2 TO-BE Process

Approved incident update -> data validation -> criticality classification -> structured alert -> owner routing -> automated reminder -> threshold-based escalation -> restoration/closure -> dashboard refresh. Exceptions such as incomplete data or unknown mappings are logged and routed to the approved manual-review process.

5.3 Pilot / Outcome Comparison

Outcome Metric	Before	After	Change
Total CMO-related incident dockets	607	494	-113 / 19% reduction
30-60 minute dockets	71	58	-13 / 18% reduction
1-2 hour dockets	161	104	-57 / 35% reduction
More than 3 hour dockets	248	202	-46 / 19% reduction

5.4 Benefits Realized

- Reduced total CMO-related incident dockets by 19% in the compared implementation period.
- Improved the visibility and consistency of incident ownership, follow-up and escalation.
- Reduced manual monitoring and reporting effort through standardized automation workflows.
- Provided operations and leadership with an actionable, dashboard-based view of critical incidents and performance.

6. CONTROL

6.1 Control Plan

Control Item	Target / Standard	Monitoring Method	Owner	Response if Out of Control
Alert processing	Eligible approved alerts processed using configured	Processing and exception logs	Automation Support	Investigate error; invoke manual

Control Item	Target / Standard	Monitoring Method	Owner	Response if Out of Control
	rules			fallback if required
Owner mapping	Valid routing for approved areas/categories	Mapping review and exceptions	Project Lead / Operations	Correct mapping; reprocess eligible alert
Follow-up and escalation	Rules apply to eligible open incidents	Open-incident and notification logs	Operations Users	Verify thresholds and send approved manual escalation
Dashboard freshness	Visible refresh timestamp and current data status	Dashboard health check	Automation Support	Resolve refresh issue; communicate data currency
CMO-related docket performance	Track total docket count and duration-band trends	Periodic KPI review	Operations Leadership	Review trend, root cause and improvement actions

6.2 Governance Cadence

Cadence	Review Activity	Participants	Output
Daily / operational	Review open critical incidents, exceptions and escalations.	Operations users, Area Managers	Action ownership and follow-up
Weekly	Review automation health, mapping issues and repeated exception patterns.	Project Lead, Automation Support	Configuration or process improvements
Periodic KPI review	Review CMO-related docket and duration trends.	Operations Leadership, Project Lead	Performance decision and corrective actions

6.3 Sustainment and Future Opportunities

- Maintain approved parsing, routing and escalation configuration with change control.
- Run periodic validation samples to identify input-format change or classification gaps.
- Keep a documented manual fallback process for platform or source downtime.
- Extend capability with approved ticketing integration, confidence scoring for ambiguous alerts and role-based dashboard views.

7. Conclusion

The DMAIC analysis shows that the key opportunity was not network fault resolution itself, but the operational process around recognizing, routing, tracking and escalating incidents. By standardizing those actions through AI-powered automation and dashboard visibility, the project achieved an evidenced 19% reduction in total CMO-related incident dockets during the compared period while creating a sustainable control framework for ongoing operational improvement.

8. Document Approval

Approver Role	Name / Signature	Date
Operations Leadership / Sponsor	______________________________ _	__________________
Project Lead	______________________________ _	__________________
Business Representative	______________________________ _	__________________